RPSC Programmer 2024 — Paper 1

Rajasthan Public Service Commission · Programmer (Paper 1)

100 questions · 100 marks · 2 sections

Question Paper

Q1–40 · Section A · Reasoning, Numerical Analysis & Current Affairs · 40 questions

Q41–100 · Section B · DBMS, Data Communication & Computer Fundamentals · 60 questions

Ye paper official exam ke baad public hui question paper aur answer key se taiyar kiya gaya hai. Practice ke liye print karein.

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Section A · Reasoning, Numerical Analysis & Current Affairs

Questions 1–40

Q1.

Based on the following information, answer the question given below -

A businessman has five articles P, Q, R, S and T.

I. P weighs double that of Q.

II. Q weighs four and half times of R.

III. R weighs half that of T.

IV. T weighs less than P but more than R. S is heavier than R.

Which one of the following is the lightest?

(1) Q
(2) S
(3) R
(4) P
(5) Question not attempted

Q2.

In a certain language, NOIDA is coded as QLLAD. How is AJMER coded in that language?

(1) DHOAC
(2) DGPBU
(3) DMOBC
(4) DNPAU
(5) Question not attempted

Q3.

Data of 500 students who took an examination of Physics, Chemistry and Mathematics is given below -

Passed in all the subjects | 176

Failed in all the subjects | 70

Failed in Physics | 185

Failed in Chemistry | 217

Failed in Mathematics | 201

Passed in Physics only | 84

Passed in Chemistry only | 52

Passed in Mathematics only | 73

How many students passed in at least one subject?

(1) 324
(2) 176
(3) 209
(4) 430
(5) Question not attempted

Q4.

Should there be a total ban on use of plastic bags?

Arguments:

I. Yes, use of plastic bags causes various environmental problems.

II. No, instead thickness of plastic should be specified which don't cause much damage to the environment.

(1) Both the arguments I & II are strong
(2) Neither argument I nor II is strong
(3) Only argument I is strong
(4) Only argument II is strong
(5) Question not attempted

Q5.

Answer the question based on the data given in the table below:

[Table showing Gender/Performance distribution with rules:

A) Half of the students are either Excellent or Good.

B) 40% of the students are females.

C) One third of the male students are average.]

Number of female students whose performance is good, is -

(1) 16
(2) 22
(3) 8
(4) 0
(5) Question not attempted

Q6.

Three consecutive integers are written in increasing order. What is the second integer?

Statements:

(i) The sum of the first and third integer is 22.

(ii) The sum of the thrice of third integer, first integer and second integer is 57.

Give answer as:

(1) Both statements (i) and (ii) together are sufficient to answer the question but neither statement alone is sufficient

(2) If statement (ii) alone is sufficient but statement (i) alone is not sufficient to solve the question

(3) If statement (i) alone is sufficient but statement (ii) alone is not sufficient to solve the question

(4) Each statement (i) and (ii) alone is sufficient to answer the question

(5) Question not attempted

Q7.

In the following series one term is missing. The missing term is -

6, 10, 18, ?, 66, 130, 258

(1) 28

(2) 32

(3) 34

(4) 50

(5) Question not attempted

Q8.

Find the signs in the given equation, which when mutually interchanged the equation becomes correct -

$9 + 5 \div 4 \times 3 - 6 = 12$

(1) + and -

(2) ÷ and -

(3) + and ×

(4) ÷ and ×

(5) Question not attempted

Q9.

The ratio of Amit's salary to Vikas's salary is 4:5. How much is Vikas's expenditure?

Statements:

(i) The ratio of Amit's and Vikas's expenditure is 5:6.

(ii) The ratio of their savings is 1:1.

Give answer as:

(1) If statement (i) and (ii) both are required to answer the question

(2) If statement (i) and (ii) both are not sufficient to answer the question. More data (information) are needed to answer the question

(3) If statement (ii) alone is sufficient to answer the question

(4) If statement (i) alone is sufficient to answer the question

(5) Question not attempted

Q10.

If January 1st, 2007 is Monday, then what was the day on 1st January, 1995?

(1) Friday
(2) Monday
(3) Saturday
(4) Sunday
(5) Question not attempted

Q11.

A statement and two conclusions (i) and (ii) are given below. You have to consider the statement as true though it seems to be different from the actual facts.

Statement: All the scientists are males. Abdul Kalam is a scientist.

Conclusions:

(i) Abdul Kalam is a male.

(ii) Females cannot be the scientists.

Which of the conclusion/s follow/s from the given statement?

(1) Neither conclusion (i) nor (ii) follows
(2) Both the conclusions (i) and (ii) follows
(3) Only conclusion (ii) follows
(4) Only conclusion (i) follows
(5) Question not attempted

Q12.

City C is in which direction to city A?

Statements:

I. C is to the North of D, which is to the West of B.

II. B is to the South - East of C.

III. D is to the North - East of A.

Which of the statement/s is/are sufficient to answer the question?

(1) Only Statement I
(2) Statements I and III
(3) Statements I and II
(4) Statements II and III
(5) Question not attempted

Q13.

Study the table and find the ratio P:R -

Article | Cost Price | Market Price | Discount | Selling Price | Profit/Loss(%)

1 | 12000 | +20% | 10% | P | M

2 | 12500 | +10% | 15% | N | Q

3 | 13000 | +30% | 20% | O | R

(1) 2:1
(2) 3:2
(3) 2:3
(4) 1:2
(5) Question not attempted

Q14.

The sum of a positive number and its reciprocal is twice the difference of the number and its reciprocal. The number is -

(1) √3
(2) 1/√3
(3) √2
(4) 1/√2
(5) Question not attempted

Q15.

The two sides of a triangle measure 5 and 12. The length of the third side of the triangle lies between the interval -

(1) 13 to 23
(2) 7 to 17
(3) 6 to 10
(4) 3 to 8
(5) Question not attempted

Q16.

In which region of the moon is 'Shiv Shakti Point' located?

(1) Near the South Pole
(2) Near the Equator
(3) Near the Sea of Tranquility
(4) Near the North Pole
(5) Question not attempted

Q17.

The 'Smart Laboratory on Clean Rivers (SLCR)' in Varanasi is being established in collaboration with which of the following country?

(1) Denmark
(2) Norway
(3) Sweden
(4) Iceland
(5) Question not attempted

Q18.

Who of the following is the author of the book "Why Bharat Matters"?

(1) Shashi Tharoor
(2) Subrahmanyam Jaishankar
(3) J Sai Deepak
(4) Bhupender Yadav
(5) Question not attempted

Q19.

Gati Shakti Vishwavidyalaya is situated in -

(1) Maharashtra
(2) Tamil Nadu
(3) Gujarat
(4) Rajasthan
(5) Question not attempted

Q20.

'Kharchi Puja' is a major festival of -

(1) Tripura
(2) Jharkhand
(3) Bihar
(4) Chhattisgarh
(5) Question not attempted

Q21.

For which of the following product French delegation visited Rajasthan in July, 2024 to study the possibility of GI tag?

(1) Kota Doria
(2) Pokaran Pottery
(3) Sojat Mehandi
(4) Mathania Chilli
(5) Question not attempted

Q22.

'National Milk Day' is celebrated every year on -

(1) 28th August
(2) 26th November
(3) 22nd September
(4) 12th December
(5) Question not attempted

Q23.

Who among the following was NOT a member of Padma Awards Committee, 2024?

(1) Jogendra Nath Phukan
(2) Satnam Singh Sandhu
(3) I.V. Subbarao
(4) Tariq Mansoor
(5) Question not attempted

Q24.

How many Delimitation Commissions have been constituted by the Government of India till July, 2024?

(1) Three
(2) Four
(3) Two
(4) One
(5) Question not attempted

Q25.

In the modified Budget 2024-25 of Rajasthan, at which place the construction of Adivasi Panorama has been announced by the Chief Minister?

(1) Banswara
(2) Dungarpur
(3) Karauli
(4) Dausa
(5) Question not attempted

Q26.

The first multi-sports beach games in India, 'The Beach Games 2024', was organized at -

(1) Kasarkod Beach, Karnataka
(2) Golden Beach, Odisha
(3) Eden Beach, Puducherry
(4) Ghoghla Beach, Diu
(5) Question not attempted

Q27.

The Indian rupee sign (₹) was adopted by the Government of India in the year -

(1) 2012
(2) 2010
(3) 2009
(4) 2011
(5) Question not attempted

Q28.

How many aerodromes are operational in India as on 1st August, 2024?

(1) 127
(2) 157
(3) 141
(4) 109
(5) Question not attempted

Q29.

The name of the project, led by UGC in collaboration with Bhartiya Bhasha Samiti, which aims to produce 22,000 books in the next five years in 22 scheduled languages is -

(1) PRAGATI
(2) ASMITA
(3) STARS
(4) BRIDGE
(5) Question not attempted

Q30.

Consider the following statements -

(I) In 2023-24, Agriculture and Allied Sectors contributed 28.21 percent in Rajasthan's Gross State Value Added at current prices.

(II) In the year 2023-24, Service Sector continued to be the largest sector in the state with a share of 45.07 percent in Rajasthan's Gross State Value Added at current prices.

Correct answer is -

(1) Only statement (I) is true
(2) Both the statements (I) and (II) are false
(3) Only statement (II) is true
(4) Both the statements (I) and (II) are true
(5) Question not attempted

Q31.

'Dhabhai Kund' is located in -

(1) Jodhpur
(2) Bundi
(3) Alwar
(4) Kota
(5) Question not attempted

Q32.

In the modified Budget 2024-25 of Rajasthan, LEAP program has been announced under i-Start to upskill start-up founders and youth and to provide them skilled manpower. Here the full form of LEAP is -

(1) Learn, Earn And Progress
(2) Learn, Excel And Progress
(3) Learn, Earn And Pay
(4) Learn, Excel And Pay
(5) Question not attempted

Q33.

Through which toll free helpline mental health consultation is being provided by psychological experts in Rajasthan?

(1) Tele-ASHA
(2) Tele-CHITTA
(3) Tele-NIDAN
(4) Tele-MANAS
(5) Question not attempted

Q34.

Under which section of the 'Bharatiya Nagarik Suraksha Sanhita 2023' is Zero FIR mentioned?

(1) Section 302
(2) Section 173
(3) Section 428
(4) Section 150
(5) Question not attempted

Q35.

Which mineral associated with Tungsten is found in the Degana area of Rajasthan?

(1) Hematite
(2) Wolframite
(3) Magnetite
(4) Bauxite
(5) Question not attempted

Q36.

Consider the following statements regarding 'Mission Madhuhari' in Rajasthan -

(A) It is started to prevent Type-1 Diabetes.

(B) This program is started under the joint aegis of the Medical and Health Department, National Health Mission and Ford Foundation.

Choose the correct option -

(1) Both the statements (A) and (B) are incorrect
(2) Only statement (A) is correct
(3) Both the statements (A) and (B) are correct
(4) Only statement (B) is correct
(5) Question not attempted

Q37.

Germplasm station of RCDF ltd. is situated at -

(1) Nadol (Pali)
(2) Barsingsar (Bikaner)
(3) Narwa Khichiyan (Jodhpur)
(4) Jobner (Jaipur)
(5) Question not attempted

Q38.

Which Jaipur film Director's debut film 'Sad letters of an imaginary woman' has received the Asian Cinema Fund 2024 award?

(1) Nidhi Saxena
(2) Monika Jain
(3) Sandeep Kaur
(4) Taru Jindal
(5) Question not attempted

Q39.

Match List-I with List-II and select the correct answer by using the code given below the lists -

List-I List-II

(Book) (Writer)

(A) Sangeet Raj (i) Krishnanand Vyas

(B) Raag Manjari (ii) Bhavbhatt

(C) Anup Vilas (iii) Pundarika Vittala

(D) Raag Kalpdrum (iv) Maharana Kumbha

Code:

(1) A-(iii), B-(ii), C-(iv), D-(i)
(2) A-(i), B-(ii), C-(iii), D-(iv)
(3) A-(iv), B-(iii), C-(ii), D-(i)
(4) A-(iv), B-(i), C-(ii), D-(iii)
(5) Question not attempted

Q40.

Hemchandra Vikramaditya, the Hindu emperor is related to which district of Rajasthan?

(1) Ajmer
(2) Jalore
(3) Sirohi
(4) Alwar
(5) Question not attempted

Section B · DBMS, Data Communication & Computer Fundamentals

Questions 41–100

Q41.

The reliable data transfer is implemented by

(1) Network layer
(2) Physical layer
(3) Presentation layer
(4) Transport layer
(5) Question not attempted

Q42.

For Stop-and-Wait ARQ, for 10 data packets sent, ... acknowledgements are needed.

(1) 9
(2) 20
(3) 11
(4) 10
(5) Question not attempted

Q43.

What does /24 denote in CIDR?

(1) 24 bits are used for host portion
(2) 24 bits are used for network portion
(3) 24 subnets are created
(4) 24 addresses are available in the subnet
(5) Question not attempted

Q44.

IP address 172.16.123.213 is found to be residing on a sub-network with the following subnet mask: 255.255.255.240. What is the network address of this subnet?

(1) 172.16.123.240
(2) 172.16.123.208
(3) 172.16.123.213
(4) 172.16.123.245
(5) Question not attempted

Q45.

Which of the following concurrency control protocols ensure both conflict serializability and freedom from deadlock?

I) 2PL (Two Phase Lock)

II) Time-Stamp ordering

III) Multiple granularity protocol

(1) III only
(2) I only
(3) II only
(4) I and II both
(5) Question not attempted

Q46.

In ER Model constructs, which type of attribute is most suitable to represent the skill(s) of an employee for an organization?

(1) Composite attribute
(2) Multivalued attribute
(3) Atomic attribute
(4) Derived attribute
(5) Question not attempted

Q47.

The size of the 'total length field' of IPv4 datagram is -

(1) 16 bits
(2) 32 bits
(3) 8 bits
(4) 4 bits
(5) Question not attempted

Q48.

Consider Relations R1 and R2 as given below -

[Relations shown in tables]

The number of tuples in output of R1 ⋈ R2 will be -

(1) 20
(2) 7
(3) 6
(4) 5
(5) Question not attempted

Q49.

In IPv4, packet fragmentation and reassembly is handled by

(1) using the More Fragments and Fragment Offset Fields
(2) using the MAC Header
(3) using the Header Checksum
(4) encapsulating packets in larger data frames
(5) Question not attempted

Q50.

What network property is achieved through the use of the Automatic Repeat reQuest (ARQ) mechanism?

(1) Reliability
(2) Scalability
(3) Availability
(4) Confidentiality
(5) Question not attempted

Q51.

What is the role of DNS? It is best described as -

(1) maps a domain name to an IP address
(2) translate a host's private IP address to a public IP address
(3) translate a host's IPv6 address to an IPv4 address
(4) maps a MAC address to an IP address
(5) Question not attempted

Q52.

The number of attribute in a relation is known as -

(1) Degree of the relation
(2) State of the relation
(3) Features of the relation
(4) Address of the relation
(5) Question not attempted

Q53.

Consider following schedule with a sequence of operations on a data object Q having the initial value of 500 -

[Table showing operations T1 and T2]

What will be the final value of Q after the last operation is completed?

(1) 700
(2) 500
(3) 550
(4) 600
(5) Question not attempted

Q54.

Consider a schema R (A, B, C, D) and functional dependencies A → B, C → D. Then, the decomposition of R into R1 and R2 where R1 (AB) and R2 (CD), is -

(1) Lossless join but not dependency preserving
(2) Dependency preserving and lossless join
(3) Not dependency preserving and not lossless join
(4) Dependency preserving but not lossless join
(5) Question not attempted

Q55.

Consider below statements -

(I) Each non-leaf node in B+ tree has between [n/2] and n Children, where n is fixed for a particular tree.

(II) The search key of a clustering index is always primary key.

(III) Secondary indices must be dense.

(IV) In a dense index, an index entry appears for only some of the search-key values.

Which of the above statements are true?

(1) I and II
(2) I and III
(3) II and III
(4) I and IV
(5) Question not attempted

Q56.

The message 1001 is to be transmitted using the CRC polynomial $x^3 + x + 1$ to protect it from errors. The message that should be transmitted is -

(1) 1001100
(2) 1001011
(3) 1001110
(4) 1001000
(5) Question not attempted

Q57.

Which operation allows us to find tuples that are in one relation but not in another relation?

(1) The Set – Difference Operation
(2) Cartesian Product Operation
(3) The Union Operation
(4) The Set Intersection Operation
(5) Question not attempted

Q58.

A Reverse Lookup in DNS means a lookup to ..... using ..... records.

(1) identify mail servers, SOA
(2) map IP address to domain name, PTR
(3) map domain names to IP address, A
(4) determine the authoritative name servers, CNAME
(5) Question not attempted

Q59.

Consider a relation with attributes {A, B, C, D} and the following functional dependencies: A → B, B → C and C → D. If A is the primary key, which of the following is true regarding BCNF?

(1) The relation is in BCNF because it is in 3NF.
(2) The relation is not in BCNF.
(3) The relation is in 3NF but not in BCNF.
(4) The relation is in BCNF because all determinants are super keys.
(5) Question not attempted

Q60.

The primary use of scramblers in Line Coding Schemes is to -

(1) make the data look random
(2) compress transmission data
(3) increase signal bandwidth
(4) decrease signal bandwidth
(5) Question not attempted

Q61.

In a slotted ALOHA with G-stations attempting to transmit in one slot, the throughput is equal to -

(1) $G \times e^{-G}$
(2) $2G \times e^{-G}$
(3) e^{-G}
(4) $G \times e^{-2G}$
(5) Question not attempted

Q62.

Two phase locking protocol has which of the following problem(s)?

(1) Chances of deadlock
(2) Cascading Rollback
(3) Neither 1 nor 2
(4) Both 1 & 2
(5) Question not attempted

Q63.

Which operation requires two relations as input to form a new relation and these two relations must have a common attribute?

(1) Projection operation
(2) Cartesian product operation
(3) Division operation
(4) Natural join operation
(5) Question not attempted

Q64.

Consider following relation S(S1, S2, S3, E1, E2) and functional dependencies hold on the schema are as -

FD1: S1 → {S2, S3}

FD2: {E1, S3, S2} → S1

FD3: E2 → {S3, S2, E1}

What is the highest normal form of the relation S?

(1) 1NF
(2) 2NF
(3) 3NF
(4) BCNF
(5) Question not attempted

Q65.

Which of the following protocol is used for transferring email messages from one machine to another?

(1) SNMP
(2) FTP
(3) TELNET
(4) SMTP
(5) Question not attempted

Q66.

Let us assume that there are four (4) network segments in Bus topologies: A1 – A2, B1 – B2 – B3, C1 – C2 and D1 – D2 – D3. If A1, B1, C1 and D1 are interconnected in mesh, then how many mesh connections are required?

(1) 12
(2) 10
(3) 6
(4) 45
(5) Question not attempted

Q67.

An organization has a Class B network and wishes to form subnets for 64 departments. The subnet mask would be -

(1) 255.255.128.0
(2) 255.255.0.0
(3) 255.255.64.0
(4) 255.255.252.0
(5) Question not attempted

Q68.

Indices whose search key specifies an order different from the sequential order of the file are called -

(1) Primary Indices
(2) Random Indices
(3) Secondary Indices
(4) Sequential Indices
(5) Question not attempted

Q69.

Consider the following instance of a relation District -

District

Name | Population

JA | 10

AJ | 50

MM | 5

Which of the District name will be returned by the following relational algebra expression?

(πName (σName="AJ"(District))) - [(πName (σPopulation>5(District))) ∩ (πName (σName="AJ"(District)))]

(1) JA
(2) MM
(3) AJ
(4) JA, AJ
(5) Question not attempted

Q70.

The cardinality of a relational table with 5000 rows and 10 columns is -

(1) 500
(2) 5000
(3) 10
(4) 50000
(5) Question not attempted

Q71.

Which of the following Multiple Access Protocols is not suitable for wired networks?

(1) CSMA/CD
(2) TGMA
(3) TDMA
(4) CSMA/CA
(5) Question not attempted

Q72.

What is the advantage of setting up a DMZ (Demilitarized Zone) using firewall?

(1) To create multiple zones of trust
(2) Load balancing on the server
(3) Improve network performance
(4) Brings internet service into home from internet service provider
(5) Question not attempted

Q73.

In ER Model, which of the following relationship is correct for a condition that a person in INDIA has an Aadhaar Card?

(1) One to one relationship
(2) Many to many relationship
(3) Many to one relationship
(4) One to many relationship
(5) Question not attempted

Q74.

Which one of the following statement is not correct?

(1) Self-describing nature of database system is one of the characteristics of the database approach.
(2) Data models provide necessary means to achieve data abstraction.
(3) Database approach doesn't support data independence between data and programs.
(4) Database approach supports multiple views of the data.
(5) Question not attempted

Q75.

In Relational Model, which type of constraint does not allow composite and multivalued attributes?

(1) Uniqueness constraints
(2) Referential integrity constraints
(3) Domain constraints
(4) Entity integrity constraints
(5) Question not attempted

Q76.

Consider following two instances of relation Student and Faculty with operation on these instances -

Student Faculty

D-Name | Strength D-Name | Strength

CS | 900 EE | 2000

IT | 500 ECE | 5000

Cyber | 300 Cyber | 300

ME | 1000 DS | 100

CIVIL | 2000 CIVIL | 2000

(a) Student - Faculty

(b) (Student ∩ Faculty) ∪ (Faculty ∩ Student)

(c) (Student ∪ Faculty) - (Student ∩ Faculty)

(d) Student ∩ Faculty

Identify the correct option that produces equivalent output -

(1) b, d

(2) a, b

(3) b, c

(4) b, c, d

(5) Question not attempted

Q77.

If a Class B network on the internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet?

(1) 2046

(2) 1022

(3) 1023

(4) 1024

(5) Question not attempted

Q78.

There are n stations in a slotted LAN. Each station attempts to transmit with a probability p in each time slot. What is the probability that only one station transmits in a given time slot?

(1) $n \times p \times (1 - p)^{n-1}$

(2) $p \times (1 - p)^{n-1}$

(3) $1 - (1 - p)^{n-1}$

(4) $(1 - p)^{n-1}$

(5) Question not attempted

Q79.

'View' in a database is -

(1) A physical table that stores data

(2) A virtual table created by querying one or more tables

(3) A method to improve database performance

(4) A backup copy of a database

(5) Question not attempted

Q80.

The three-way handshake mechanism in TCP aims to -

(1) secure the communication
(2) send bits from one device to another over an optical fibre connection
(3) perform error detection and correction
(4) establish connection and synchronize sequence numbers
(5) Question not attempted

Q81.

Total time to access a disk block is given by the -

(1) Seek time + transfer time
(2) Seek time + rotational delay + transfer time
(3) Seek time + rotational delay
(4) Seek time
(5) Question not attempted

Q82.

Which set of SQL language commands is used to query and update a database?

(1) Data Manipulation Language (DML)
(2) Data Control Language (DCL)
(3) Data Definition Language (DDL)
(4) Both Data Control Language (DCL) and Data Definition Language (DDL)
(5) Question not attempted

Q83.

The Address Resolution Protocol (ARP) is used for -

(1) finding the IP address that corresponds to a MAC address
(2) finding the MAC address that corresponds to an IP address
(3) finding the IP address from the DNS
(4) finding the IP address of the default gateway
(5) Question not attempted

Q84.

An ER Model describes which type of schema?

(1) Physical Schema
(2) Internal Schema
(3) External Schema
(4) Conceptual Schema
(5) Question not attempted

Q85.

The operation which can be used to specify automatic actions that the database system will perform when certain events and conditions occur is known as -

(1) CREATE TRIGGER
(2) CREATE ASSERTION
(3) CREATE VIEW
(4) CREATE TABLE
(5) Question not attempted

Q86.

Correct order of applying 3 passes in ARIES recovery algorithm is -

(1) Undo, Analysis, Redo
(2) Analysis, Undo, Redo
(3) Analysis, Redo, Undo
(4) Undo, Redo, Analysis
(5) Question not attempted

Q87.

Which of the following protocol may be used as zeroconf and plug-and-play?

(1) TCP
(2) DHCP
(3) NAT
(4) ICMP
(5) Question not attempted

Q88.

How many bits for 'Fragment Offset' field are there in IPv4 header?

(1) 16 bits
(2) 10 bits
(3) 13 bits
(4) 32 bits
(5) Question not attempted

Q89.

Consider a relation R with attributes (A, B, C), where B is the only candidate key. Identify the total number of possible super keys of the relation R.

(1) 2
(2) 4
(3) 3
(4) 1
(5) Question not attempted

Q90.

Any relation that is not a part of the Logical Model, but is made visible to a user as a virtual relation is called a -

(1) View
(2) Instance
(3) Schema
(4) Tuple
(5) Question not attempted

Q91.

Consider following two instances of relations X and Y -

X Y

Code | Item Item | Quantity

100 | A A | 5

200 | B B | 6

300 | B C | 3

400 | C

How many total tuples a Cartesian product between the two relations X and Y will produce?

(1) 3
(2) 7
(3) 4
(4) 12
(5) Question not attempted

Q92.

The size of MAC address (in bits) is -

(1) 42
(2) 48
(3) 24
(4) 36
(5) Question not attempted

Q93.

Network Allocation Vector (NAV) is associated with which of the following MAC technologies?

(1) Wi-Fi
(2) CSMA/CD
(3) Bluetooth
(4) Ethernet
(5) Question not attempted

Q94.

ICMP protocol is a ..... layer protocol.

(1) Data link
(2) Transport
(3) Application
(4) Network
(5) Question not attempted

Q95.

Which SQL statement is used in embedded SQL to fetch data into host variables?

(1) INSERT INTO
(2) DELETE
(3) SELECT INTO
(4) UPDATE
(5) Question not attempted

Q96.

Which of the following is true for Circuit Switching handling multiple simultaneous connections?

(1) A single path is used and shared for all data
(2) All connections are multiplexed into a single channel
(3) It encodes all data into a single channel
(4) Separate physical paths are allocated for each channel
(5) Question not attempted

Q97.

Which of the following is not related to data link layer?

(1) Error control
(2) Congestion control
(3) Framing
(4) MAC addressing
(5) Question not attempted

Q98.

What are the maximum number of spanning trees (Tn) possible for a complete graph K5 on 5 vertices?

(1) 20
(2) 125
(3) 100
(4) 200
(5) Question not attempted

Q99.

An organized collection of logically related data is called -

(1) Data type
(2) Database
(3) Metadata
(4) Transaction
(5) Question not attempted

Q100.

The physical layer is responsible for -

(1) Bit-by-bit delivery
(2) Port to port delivery
(3) Process to process delivery
(4) Application to application delivery
(5) Question not attempted